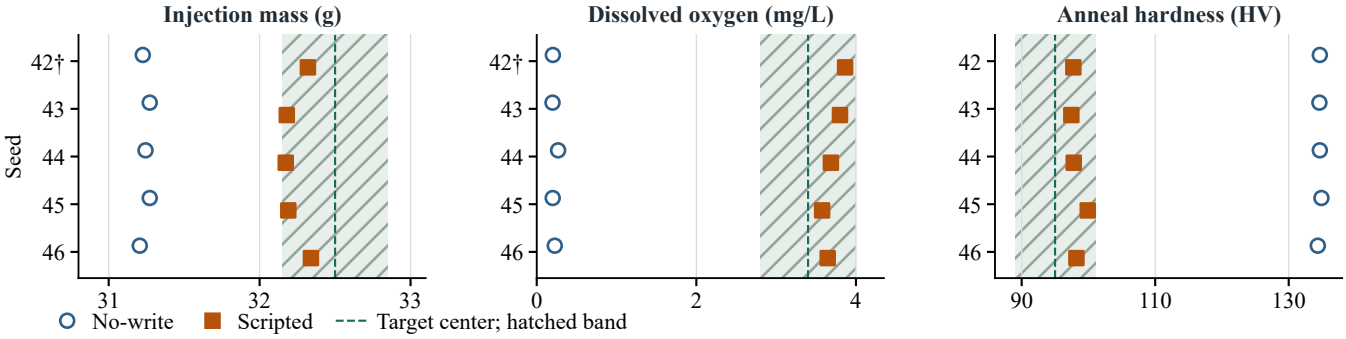
(a) Simulator endpoints

No-write: 60 s; mean of final five samples

Scripted: endpoint after scenario-specific steps



† No-write seed 42 in injection and wastewater began from residual states.

(b) Process-window ablation

	Repetition 1		Repetition 2		Repetition 3		
	Below	Above	Below	Above	Below	Above	HTTP 200
Full guard	○	○	○	○	○	○	0/6
Window check removed	■	■	■	■	■	■	6/6
Readback check removed	○	○	○	○	○	○	0/6
Window + readback removed	■	■	■	■	■	■	6/6

Open circle: HTTP 400 (rejected) | Filled square: HTTP 200